Functional Design Document

Project BattleBot

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Introduction

Background

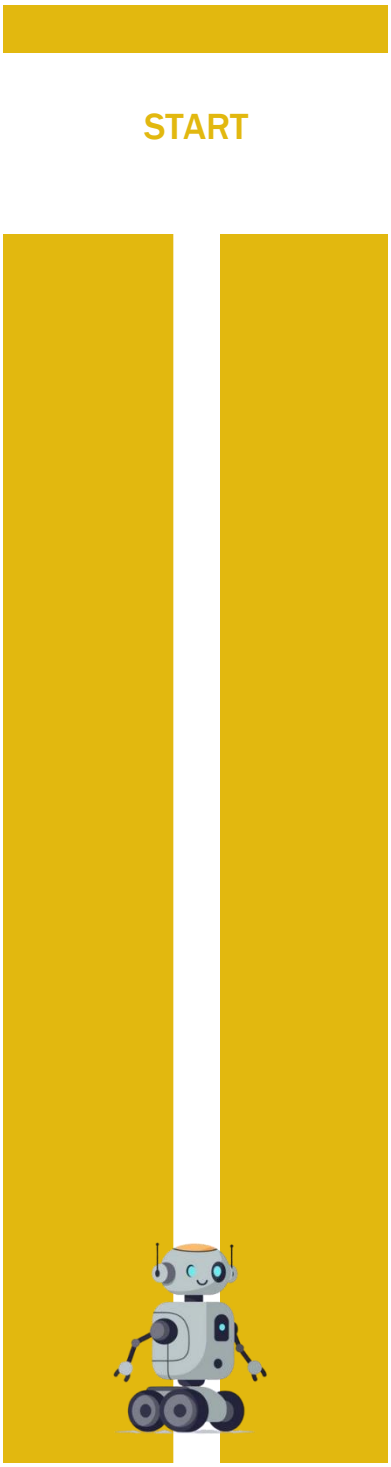
The BattleBot Relay Project consists of a three-stage autonomous robotics challenges.

The goal is to design a relay system where three robot subsystems cooperate to complete a race track consisting of:

- A line following section
- A straight line maze
- A maze section

Each subsystem is developed by a subgroup of two students. Together, the six students form a single integrated relay system.

The robots must operate autonomously and transfer a physical object (small pillar) between stages without human intervention.



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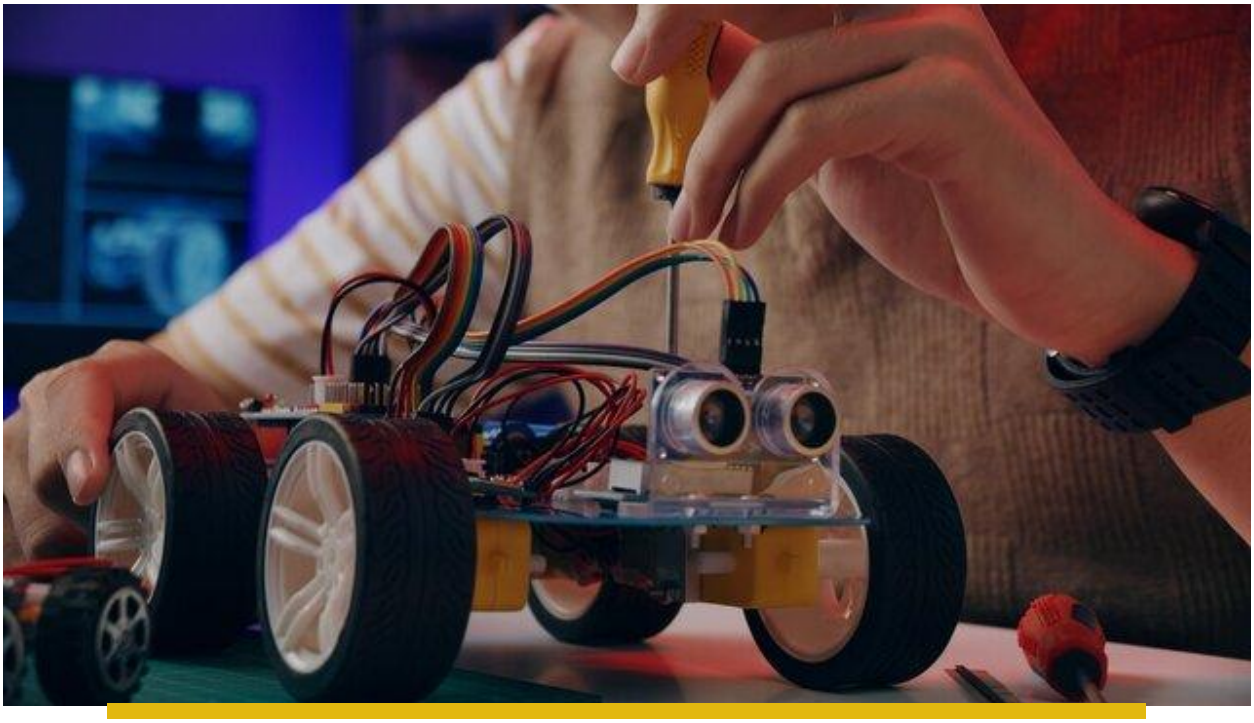


Objective

The objective of the project is to design, implement, and integrate three autonomous robotic subsystems that:

- Complete their assigned track section.
- Accurately stop at the designated transfer points.
- Transfer the pillar reliably to the next subsystem.
- Complete the full relay without manual assistance.

Educational Objectives

In addition to completing the relay race, the project aims to develop practical knowledge of embedded systems engineering.

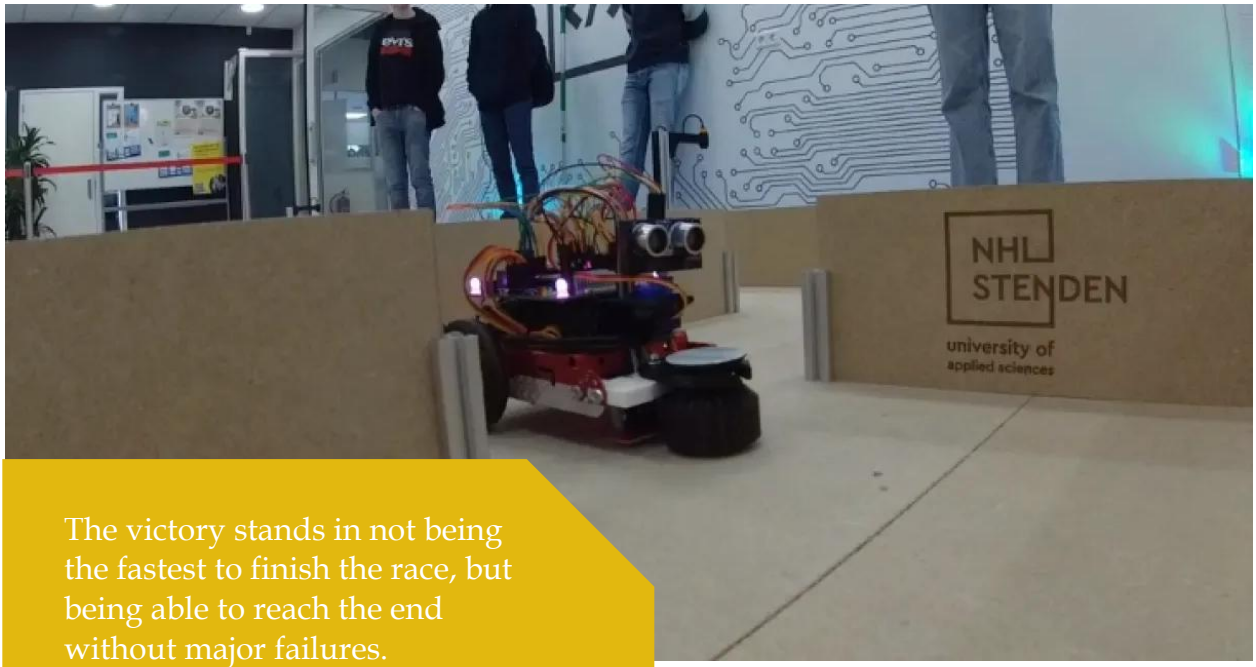
Throughout the project, weekly programming assignments were completed to progressively build understanding in:

- Digital input and output control
- Timing and delay management
- State-based programming logic
- Motor control using H-bridge drivers
- Sensor integration (line sensors, ultrasonic sensors, encoders)
- Modular programming using functions

These assignments provided the technical foundation necessary for the development of the final relay system.

System Overview

1. First team grabs the object after race flag signals beginning of the race
2. Robot follow black circular line
3. During the track the robot needs to be prepared to avoid objects
4. Stops at the designated drop-off zone
5. Transfers the object
6. Second team robot retrieves the object
7. Starts to follow the straight-line maze track
8. Detects any objects and avoids it
9. Completes the line maze, successfully reaching the drop-off zone
10. Drops object
11. Third team robot picks the object
12. Starts the maze
13. Completes the race by reaching the end of the maze and delivering the object on last destination

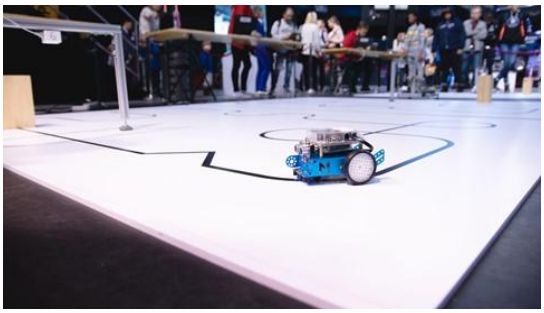


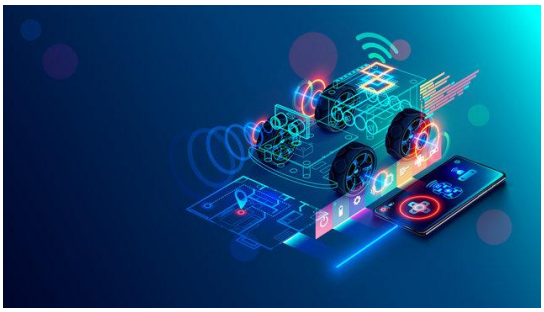
The victory stands in not being the fastest to finish the race, but being able to reach the end without major failures.

Functional Requirements

FR1 – Autonomous Operations
The relay system shall operate without manual intervention.

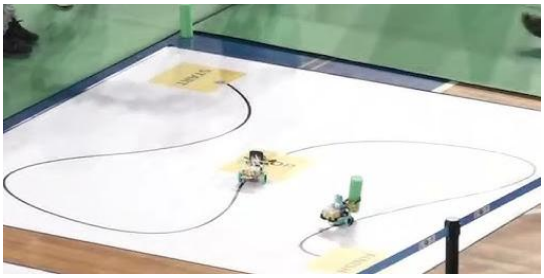
FR2 – Line Following
The straight-line and circular robots shall follow black track lines

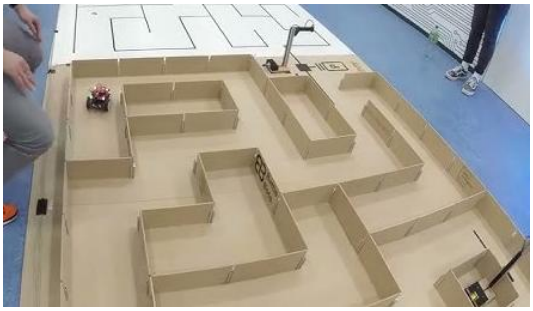




FR3 – Relay Transfer
Each Subsystem shall transfer the pillar safely to the next subsystem.

FR4 -Synchronization
Robots shall detect the correct stopping location before handover





FR5 – Maze Completion
The maze subsystem shall navigate from entrance to exit autonomously

Subsystem Requirements

Straight-Line Subsystem:

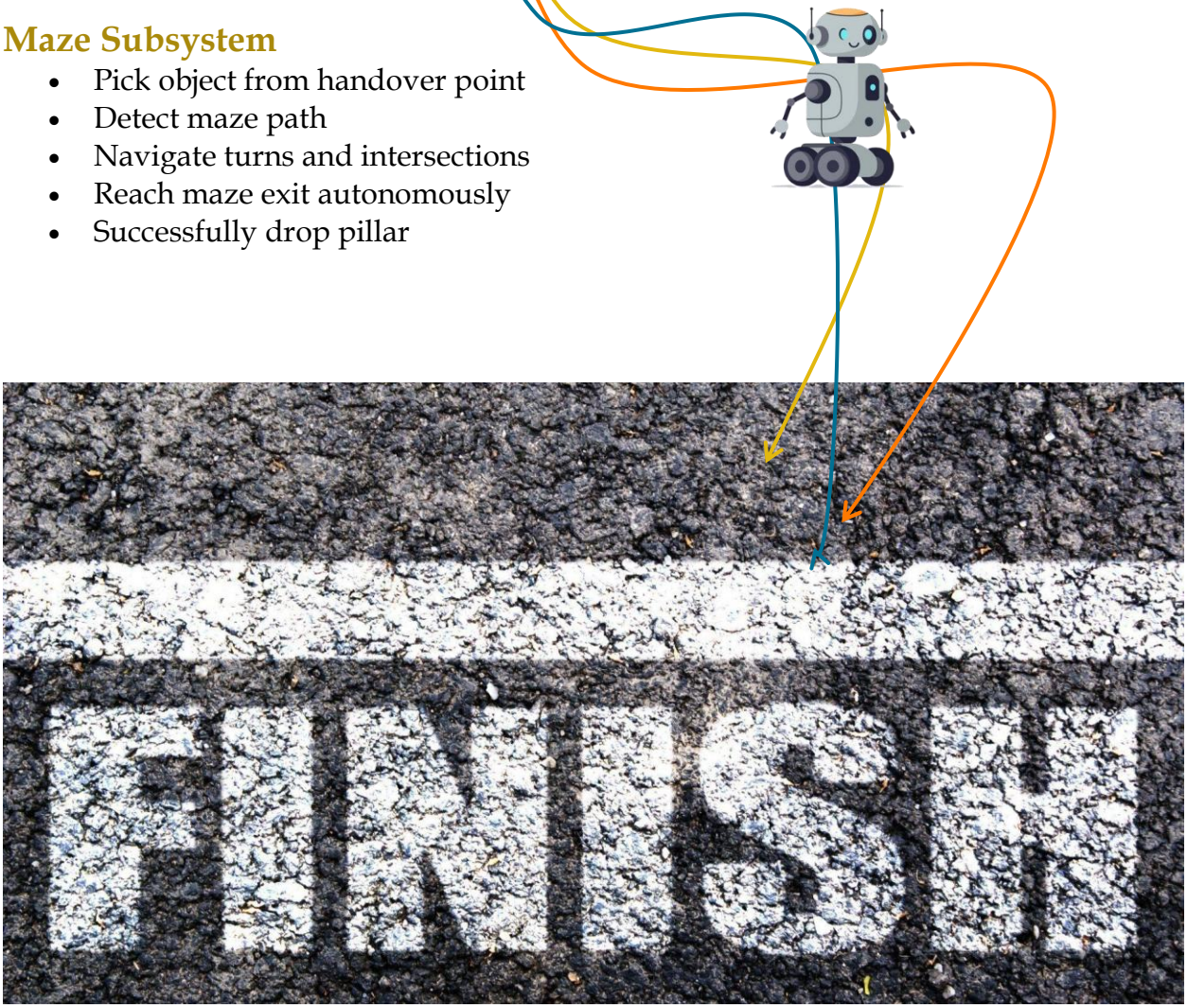
- Pick the object with the grippers successfully
- Track straight black line
- Maintain directional stability
- Stop at handover zone
- Securely transfer pillar

Circular Line Subsystem

- Pick object/pillar from handover point
- Follow circular black track
- Detect maze entrance position
- Stop precisely
- Release pillar
- Maintain orientation for handover

Maze Subsystem

- Pick object from handover point
- Detect maze path
- Navigate turns and intersections
- Reach maze exit autonomously
- Successfully drop pillar



Non-Functional Requirements

Performance

- The complete relay shall be executed within the time limit defined during testing.
- Each subsystem shall respond to sensor input within 200 ms.
- The robot shall detect obstacles within a range of 5–30 cm (where applicable).

Accuracy

- Straight and circular subsystems shall maintain line tracking with minimal deviation.
- Distance measurements based on wheel encoders shall have a maximum error of ±5%.
- 90° turns shall be accurate within ±10°.

Reliability

- The system shall complete at least three consecutive full relay runs without hardware failure.
- No manual intervention shall be required during execution.

Safety

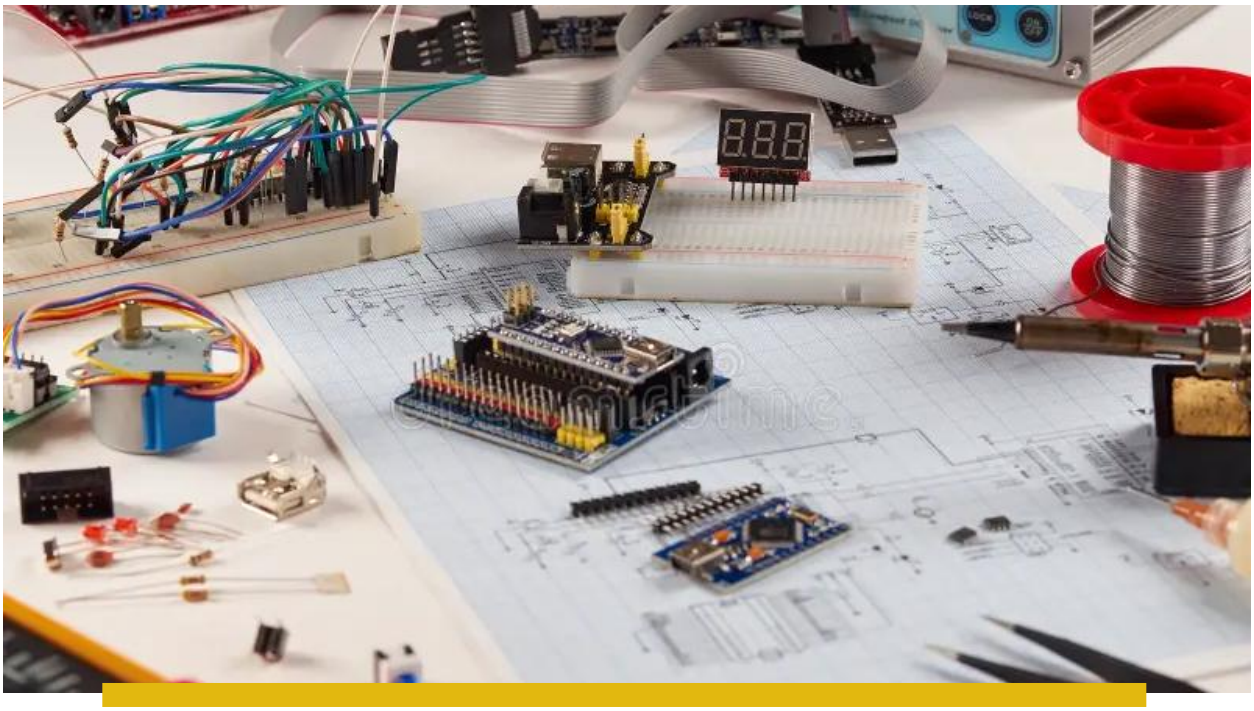
- Motors shall stop when no control signal is present.
- The robot shall not operate at voltages exceeding component specifications.
- Mechanical handover of the pillar shall not damage hardware.

Maintainability

- The software shall be modular and divided into functions.
- Pin configurations shall be clearly documented.
- Code shall follow structured programming principles.

Hardware Constraints

- The system shall operate on Arduino Nano hardware.
- Memory usage must remain within microcontroller limits.
- All components shall operate at 5V logic level



Constraints

Hardware Limitations

The system is built using Arduino Nano microcontrollers, which provide limited SRAM and flash memory. This restricts the complexity of algorithms and requires efficient programming practices.

Sensor Limitations

Ultrasonic sensors may produce unstable readings due to reflection angles and environmental interference. Line sensors may be affected by lighting conditions.

Mechanical Constrains

Wheel slippage and uneven surfaces may affect movement precision and encoder accuracy.

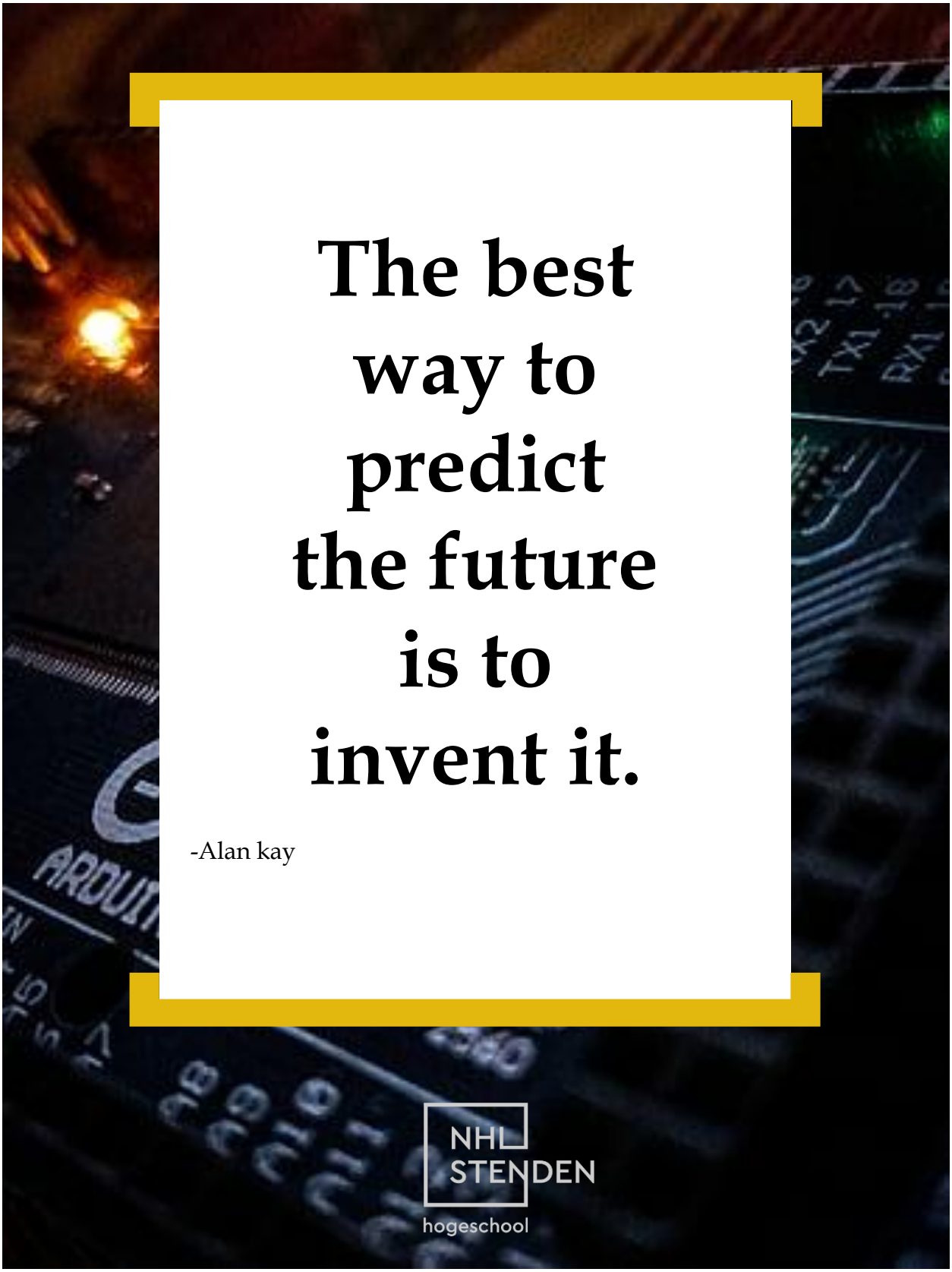
Power Supply

Battery voltage fluctuations may influence motor speed and sensor stability.

Acceptance Criteria

The system will be considered successful if:

1. The circular line subsystem completes its section without leaving the track.
2. The pillar is successfully transferred to the straight-line maze subsystem without manual intervention.
3. The straight-line maze subsystem follows the track and stops at new drop off zone for the maze race.
4. The maze subsystem completes the maze without external assistance.
5. The relay can be executed without any major mechanical or software failure
6. All subsystems operate autonomously after activation.


The best way to predict the future is to invent it.
-Alan kay
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